

A Systematic Literature Review of LLM-Based Linguistic Steganography

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This systematic literature review examines how Large Language Models (LLMs) are reshaping linguistic steganography. Through analysis of 25 primary studies, the review synthesizes recent work on generative, rewriting-based, black-box, and context-aware approaches, with particular attention to application domains, evaluation practices, external knowledge integration, and recurring technical trade-offs. The surveyed literature suggests that LLM-based methods can improve text fluency and contextual plausibility relative to earlier linguistic steganography techniques, but it also shows that gains in payload, naturalness, and security remain difficult to balance in practice. A recurring theme across the reviewed studies is the growing importance of contextual compatibility, including domain adaptation, retrieval, and semantic guidance, as a mechanism for improving both usability and concealment. The review concludes by identifying unresolved challenges in evaluation standardization, robustness, tokenization reliability, and practical deployment.

Additional Key Words and Phrases: Systematic Literature Review, Linguistic Steganography, Large Language Models, LLMs, Natural Language Processing, NLP, Black-box Steganography, Context Retrieval, Generative Text Steganography, Imperceptibility

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1 INTRODUCTION

Linguistic steganography hides secrets inside ordinary sentences, but natural language offers limited redundancy for concealment [1, 2]. A single awkward synonym, statistically unusual phrase, or contextually implausible sentence can expose hidden communication. Classical approaches based on lexical or syntactic modification therefore tend to provide low payloads and leave detectable artifacts that modern steganalysis can exploit [3].

Large language models have changed this landscape by making high-quality text generation widely accessible. Their fluency, contextual sensitivity, and broad domain coverage have enabled a new generation of steganographic methods that rely on text generation, paraphrasing, retrieval, and prompting rather than only local word substitution.

These advances do not eliminate the core trade-offs of the field. Increasing embedding capacity can still degrade fluency, coherence, or statistical naturalness, while optimizing for stealth may reduce throughput or robustness. This tension is often described as the Perceptual-Statistical Imperceptibility Conflict (PSIC) [1]. Against this background, a systematic synthesis is needed to clarify how recent work defines the problem, evaluates success, and balances practical constraints.

This review addresses that need by identifying and synthesizing recent primary studies on LLM-based linguistic steganography. Rather than arguing that a single method has resolved the field’s long-standing challenges, the paper maps the emerging design space, highlights recurring evaluation practices, and identifies the main limitations that remain unresolved.

The rest of the paper is structured as follows. Section 2 introduces the background on steganography, its evaluation criteria and the capabilities of LLMs and their role in generative linguistic steganography. Section 3 reviews related

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surveys to position the present study. Section 4 outlines the review methodology, including the research questions, search strategy, and selection criteria. Section 5 presents the findings across five research questions (RQ1–RQ5). Section 6 discusses the implications and limitations of the review. Section 7 concludes the paper and outlines future research directions.

2 BACKGROUND

Information security systems broadly encompass **encryption**, **privacy**, and **concealment**, the last of which—known as **steganography**—is the focus of this review. While encryption and privacy protect message content, they do not conceal the existence of communication, which may itself arouse suspicion. Steganography instead prioritizes **imperceptibility**: embedding information into ordinary carriers (e.g., images or text) so that hidden messages remain unnoticed.

Text is a particularly challenging carrier due to its low redundancy and strict semantic constraints. The classical “Prisoners’ Problem” [4] illustrates the goal: two parties, Alice and Bob, must exchange hidden information without alerting a watchful adversary.

Textual steganography methods are typically divided into **format-based** approaches, which exploit layout or structural features, and **content-based** approaches, which modify linguistic form. Within the latter, early techniques such as **synonym substitution** embed bits by altering lexical choices, but suffer from low capacity and high detectability. More formally, **linguistic steganography** refers to concealing information in natural language by modifying or generating text while preserving fluency and meaning [5].

Traditional linguistic approaches offer limited embedding capacity and often leave statistical artifacts. Advances in deep learning and **Large Language Models (LLMs)** now enable generative methods that achieve higher text quality and more secure embedding. Evaluating such systems requires several dimensions of imperceptibility: **perceptual** (human naturalness), **statistical** (distributional similarity to natural text), and **cognitive** (semantic and contextual fidelity) [6].

A deeper theoretical perspective introduces **channel entropy**, which quantifies the information-carrying capacity of a given communication channel. Entropy sets the upper bound for embedding rates: higher entropy allows more hidden information without detection, while lower entropy restricts capacity. Achieving this bound securely requires **perfect samplers**, which can generate text indistinguishable from genuine distributional samples. These concepts underpin the design of provably secure steganographic systems.

However, LLMs [7] introduce new challenges. Their tendency toward **hallucinations** can create detectable artifacts, highlighting the **PSIC effect** (Perceptual-Statistical Imperceptibility Conflict) [1], where optimizing for perceptual fluency may undermine statistical security. Model access further shapes practical steganography: **black-box access** (e.g., commercial APIs or hosted open-weight models) offers significant advantages, delivering substantially better text quality through access to state-of-the-art models, faster generation speeds via optimized infrastructure, and minimal local resource requirements, enabling scalable deployment without the computational overhead of training or hosting large models locally. The primary trade-off is limited control and reduced transparency over internal sampling probabilities. In contrast, **white-box access** enables fine-grained control over parameters and sampling, supporting stronger security guarantees, but typically demands substantial computational resources, engineering effort, and higher latency, raising deployment barriers. This trade-off is central to evaluating the robustness and applicability of modern linguistic steganography.

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2.1 Capabilities and Approximating Natural Communication

Large Language Models (LLMs) are autoregressive, generative systems based on the Transformer architecture [8] that approximate high-dimensional distributions over natural-language sequences [2][9]. Given a prefix, an LLM emits a probability vector over the vocabulary; the next token is sampled from this vector and appended to the prefix, and the process repeats until a stopping criterion is met. During pre-training, billions of parameters are tuned on large text corpora so that the model's predictive distribution converges to the empirical distribution of the data [10]. As a consequence, modern LLMs routinely produce text whose fluency, coherence and style are indistinguishable from human writing [11]. The learned latent representations capture stylistic and semantic regularities that generalize across domains, enabling applications requiring nuanced linguistic mimicry [12].

2.2 Role in Generative Linguistic Steganography

LLMs are considered **favorable for generative text steganography** due to their ability to generate high-quality text. Researchers propose using generative models as steganographic samplers to embed messages into realistic communication distributions, such as text. This approach marks a departure from prior steganographic work, motivated by the public availability of high-quality models and significant efficiency gains.

LLMs like GPT-2 [9], LLaMA [13], and Baichuan2 [14] are commonly used as basic generative models for steganography. Existing methods often utilize a language model and steganographic mapping, where secret messages are embedded by establishing a mapping between binary bits and the sampling probability of words within the training vocabulary. However, traditional "white-box" methods necessitate sharing the exact language model and training vocabulary, which limits fluency, logic, and diversity compared to natural texts generated by modern black-box LLM APIs or hosted open-weight models. **Black-box approaches provide substantial advantages:** they deliver significantly better text quality by leveraging state-of-the-art hosted models, achieve faster generation speeds through optimized cloud infrastructure, and require minimal local resource requirements without the computational overhead of running large models locally. In contrast, white-box methods require running large models locally, increasing latency and resource consumption. These methods further inevitably alter the sampling probability distribution, thereby posing security risks [15].

New approaches, such as LLM-Stega [15], explore **black-box generative text steganography using the user interfaces (UIs) of LLMs**. This circumvents the requirement to access internal sampling distributions. The method constructs a keyword set and employs an encrypted steganographic mapping for embedding. It proposes an optimization mechanism based on reject sampling for accurate extraction and rich semantics [15].

Another framework, Co-Stega, leverages LLMs to address the challenge of low capacity in social media. It expands the text space for hiding messages through context retrieval and **increases the generated text's entropy via specific prompts** to enhance embedding capacity. This approach also aims to maintain text quality, fluency, and relevance [16].

The concept of **zero-shot linguistic steganography** with LLMs utilizes in-context learning, where samples of cocontext are used as context to generate more intelligible stegotext using a question-answer (QA) paradigm [17]. LLMs are also employed in approaches like ALiSa, which directly conceals token-level secret messages in seemingly natural steganographic text generated by off-the-shelf BERT [18] models equipped with Gibbs sampling [19].

The increasing popularity of deep generative models has made it feasible for provably secure steganography to be applied in real-world scenarios, as they fulfill requirements for perfect samplers and explicit data distributions (see Section 2) [2, 20, 21].

LLM-based steganographic methods are typically evaluated on two primary axes: imperceptibility (perceptual, statistical, and cognitive measures) and embedding capacity (e.g., bits per token or bits per word). Imperceptibility evaluations may include automatic metrics (PPL, Distinct-n, MAUVE, KL/JSD) as well as human judgements; embedding capacity is usually reported as bits/token or overall embedding rate.

3 RELATED REVIEWS

Majeed et al. (2021) [22] surveyed pre-LLM text steganography techniques, predating the current transformer era. Setiadi et al. (2025) [23] recognizes that LLMs have "revitalized" linguistic steganography, examining recent methods (2021-2025) using GPT-2 [24], GPT-3 [25], LLaMA2 [26], and Baichuan2 [27]. However, their review remains a critical examination rather than a systematic survey, leaving several key papers unaddressed. Crucially, the field has evolved from "statistical vector embedding" (Word2Vec, GloVe) to "language-model vector embedding" that exploits BERT-scale transformers and higher-dimensional semantic spaces.

This creates a methodological gap: no systematic review comprehensively maps how large-scale transformers have redefined text steganography. Modern advances extend beyond naive generation to sophisticated Controllable Text Generation (CTG) frameworks [28]. These employ Variational Autoencoders (VAEs) to model latent features and Diffusion Models to inject randomness, mitigating spurious associations between secrets and control conditions. Classical surveys emphasized synonym replacement, spacing manipulation, and Huffman coding [22]-techniques that predated LLMs. Earlier methods relied on context-free grammars (CFGs) or Markov chains, often producing syntactically correct but semantically incoherent cover texts. Contemporary approaches leverage prompt learning and prefix tuning, enabling efficient model customization without costly full fine-tuning.

Defensive strategies must evolve accordingly. Traditional steganalysis, premised on hand-crafted statistical features, falters against generative steganography's high statistical concealment. Current research must confront "stegomalware"-attacks that conceal command-and-control communications within innocuous digital media.

4 RESEARCH METHOD

This study was undertaken as a systematic literature review using the guidelines presented in Petersen et al. [29]. The goal of this review is to identify, categorize, and analyze existing literature published between 2018 and 2025 and use syntactic and semantics aspects to represent context handling in linguistic steganographic methods.

4.1 Planning

In this section, we define our research questions, the search strategy we use, and the inclusion and exclusion criteria considered to filter the results.

4.1.1 Research Questions. This systematic literature review is guided by five research questions:

- **RQ1:** What is the current state of published literature on LLM-based steganography?
- **RQ2:** What applications are being explored for LLM-based steganographic techniques?
- **RQ3:** What evaluation metrics and methods are used to assess LLM-based steganography?
- **RQ4:** How are external knowledge sources integrated into LLM-based steganographic systems?
- **RQ5:** What limitations and trade-offs characterize current LLM-based steganographic techniques?

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4.1.2 Search Strategies. The initial literature search employed a specific query string: '(steganography or watermark or "Information Hiding") and ("Large Language Model" or LLM or BERT or LAMA or GPT)'. The query was intentionally broad to maximize recall during the identification stage, even though the final synthesis focuses on LLM-based linguistic steganography. It was executed across several digital libraries, including ACM Digital Library, IEEE Digital Library, Science@Direct, Scopus, and Springer Link, to ensure broad coverage. To complement this automated search and identify additional relevant studies, a snowballing technique was also applied. This involved examining the reference lists of included studies. While snowballing primarily yielded older steganographic techniques not explicitly mentioning LLMs, these papers often utilized similar methodological approaches to contemporary LLM-based steganography, providing valuable contextual information.

4.1.3 Inclusion and Exclusion Criteria. To ensure the selection of high-quality and relevant studies, the following criteria were applied.

Inclusion Criteria Studies were included if they:

IC1: Provided full-text access.

IC2: Were published in English from 2018 onwards.

IC3: Appeared in peer-reviewed journals, conferences, or workshops.

IC4: Directly addressed linguistic steganography or closely related information-hiding techniques involving or significantly impacted by LLMs, BERT, LAMA, or GPT architectures.

IC5: Represented empirical studies, surveys, reviews, or theoretical contributions.

Exclusion Criteria Studies were excluded if they:

EC1: Were duplicates (retaining the most complete or recent version).

EC2: Were incomplete, abstract-only, or irrelevant to steganography with LLMs.

EC3: Were non-English publications.

EC4: Came from non-peer-reviewed sources (e.g., preprints, dissertations, theses, books, book chapters), unless extended from peer-reviewed conference papers.

4.2 Conducting the Search

The initial automated search across the selected digital libraries yielded a total of 1043 candidate papers. The distribution by source was: ACM Digital Library (346), IEEE Digital Library (61), Science@Direct (209), Scopus (151), and Springer Link (276). Consistent with the scope of this review, only studies published between 2018 and 2025 were considered during screening and selection. Duplicate papers were eliminated using the Parsifal tool¹ before the remaining papers underwent a multi-stage filtering process based on title, abstract, and full-text review, guided by the predefined inclusion and exclusion criteria. After title and abstract filtering, 58 papers remained. Of these, 26 were accepted for analysis with readily available PDFs, while 6 additional papers were identified as potentially relevant but were still pending PDF acquisition at the time of analysis.

4.3 Data Extraction and Classification

A Data Extraction Form (DEF) was developed to systematically collect data from each primary study to address our research questions. The form is designed in a table format consisting of the following types of information:

¹<https://parsifal>

- Bibliometric Information: paper title, type (Steganography or Watermarking), author(s), publication year, and publication venue.
- Model Details: input and output formats, key characteristics, approach classification (three-term categorical), specific LLM used (if applicable), embedding process description, and code availability.
- Datasets: all datasets employed, including their sizes.
- Context Awareness: whether the method is "Explicit," "Implicit," or "No," the context keyword (e.g., "Social Media," "Formal Document"), how context is represented (e.g., "Text," "Pretext," "Graph," "Vector"), and how it is utilized in the method.
- Evaluation Details: evaluation metrics, steganalysis models used, and the best numerical results for each reported metric.
- Strengths and Limitations: main strengths and weaknesses of the approach or model.

Following data extraction, studies were classified based on predefined categories derived from the research questions to identify trends, patterns, and gaps in the literature. The results are summarized using tables and descriptive statistics. Each research question is addressed individually with interpretation of findings and identification of future research directions.

5 RESULTS

This section presents the synthesized findings from our systematic literature review of 25 primary studies on LLM-based steganography. The results are organized around five research questions to provide a comprehensive analysis of the current state, applications, evaluation methods, knowledge integration, and limitations in this rapidly evolving field.

5.1 State of Published Literature on LLM-based Steganography (RQ1)

5.1.1 Publication Trends and Distribution. Our analysis indicates a clear rise in LLM-based steganography research after 2023, with most of the reviewed studies appearing in 2024–2025. This increase coincides with the rapid diffusion of large-scale language models in both proprietary and open-weight forms, including widely used families such as GPT, GPT-2, and LLaMA [24–26]. Across the reviewed corpus, open-weight and research-accessible models appear more frequently than closed commercial systems, reflecting the practical importance of reproducibility and controllable experimentation.

The field has also evolved from earlier white-box token-sampling schemes toward a broader design space that includes hybrid and black-box methods. The reviewed studies now span generative schemes that write stego text from scratch [1, 2, 20, 30], rewriting systems that paraphrase existing covers [31], black-box pipelines that treat the model as an opaque API [15, 32], zero-shot protocols driven primarily by prompting [17], context-aware frameworks that retrieve external information to increase semantic plausibility or payload [16, 33], and methods that emphasize stronger security guarantees [2, 20].

Year	2020	2021-2022	2023	2024-2025	Total
Publications	2	3	5	15	25

Table 1. Publication trends by year

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Model Type (Studies)	Models and Representative Works
Open-weight Models (20/25, 80.0%)	GPT-2 [2, 20, 33, 34], LLaMA/LLaMA2 [16, 17, 21, 35], BERT [6, 19, 28, 36–41], OPT [42], BART [31, 41], Qwen [43]
Proprietary Models (4/25, 16.0%)	GPT-3.5/4, ChatGPT [15, 32, 44, 45]
Custom or Task-Specific Architectures (4/25, 16.0%)	From-scratch or task-specific models [1, 38, 39, 42]

Table 2. Representative model families across the 25 studies explicitly enumerated in the current synthesis tables. Categories are non-exclusive, so totals exceed 100%.

5.2 Applications of LLM-based Steganographic Techniques (RQ2)

The reviewed studies show that **covert communication** remains the dominant application of LLM-based linguistic steganography. In this setting, secret information is embedded into seemingly benign text so that the message can pass through ordinary communication channels with minimal suspicion. LLM-based methods improve this use case by producing more fluent and contextually plausible outputs than earlier rule-based or local-substitution systems [1, 2, 15].

Within this application space, the reviewed papers explore several operational variants. Some methods generate stego text from scratch using a shared model or sampling strategy [2, 20]. Others rely on paraphrasing or rewriting so that a pre-existing cover text can be transformed while preserving its apparent meaning [31]. Recent black-box approaches such as **LLM-Stega** show that covert communication can also be implemented through commercial or hosted interfaces without direct access to token probabilities, using prompt design, keyword sets, and reject-sampling style verification instead [15]. Context-aware systems such as **Co-Stega** and **Hi-Stega** further adapt the generated text to specific domains, topics, or social settings in order to improve plausibility and increase effective payload [16, 33].

Although adjacent topics such as watermarking, authorship attribution, or prompt-hiding attacks are relevant to the broader landscape of information hiding, they were not the primary focus of the final reviewed corpus. For that reason, this review treats them as related areas rather than as central application categories within the included LLM-based steganography studies.

5.3 Evaluation Metrics and Methods (RQ3)

Performance evaluation for LLM-based steganography relies on three key categories of metrics, with significant variation in reporting standards across studies. The analysis reveals both the diversity of evaluation approaches and the need for standardization.

Metric Type	Imperceptibility	Capacity	Security	Usage
Perceptual	PPL: 3-300	BPW: 0.5-6.0	Detection: 50-98%	85%
Statistical	KLD: 0-3.3	BPT: 1.0-5.8	F1: 0.5-0.99	70%
Semantic	BLEU: 0.3-0.9	ER: 0.2-0.4	Acc: 0.5-0.99	60%
Human Eval	MAUVE: 0.2-0.9	-	-	25%

Table 3. Evaluation metrics usage and typical ranges across studies

5.3.1 Perplexity (PPL). An imperceptibility metric [46] that measures fluency, with lower values indicating better naturalness. It is recognized as a sensitive and unreliable metric for language model evaluation due to several intrinsic limitations. First, it suffers from a "confidently wrong" problem: as Baeldung, et al. [47] notes, perplexity measures only internal consistency, allowing models to assign low perplexity to grammatically perfect but factually absurd statements like "The cat is on the ceiling," since it cannot assess truth or logic. Second, it exhibits a short-text bias as Fang, et al.

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[48] demonstrated that perplexity scores are artificially inflated for short sequences despite potentially higher fluency, making it an "unqualified referee" for fair evaluation. Third, comparability across models is impossible without identical tokenization, vocabulary size directly scales perplexity - a model with fewer tokens appears deceptively better [49]. Fourth, perplexity fails to capture long-range dependencies in modern LLMs; Fang, et al. [48] argue that averaging log-likelihood across all tokens obscures performance on crucial "key tokens" by favoring predictable filler words. Finally, the metric is easily gamed through repetition, Wang, et al. [47] finds that "perplexity cannot distinguish between right emphasis and abnormal repetition," rewarding redundant text with artificially low scores. These flaws-sensitivity to length, architectural incompatibility, semantic blindness, and exploitability-collectively render perplexity an inadequate benchmark for steganographic text quality assessment.

5.3.2 MAUVE. Another imperceptibility metric that Evaluates distributional similarity between generated and reference text by quantifying the gap between neural and human-authored text using divergence frontiers. While MAUVE provides a theoretically elegant way to measure distributional gaps between generated and reference text, it remains curiously underused-appearing in just 3 of 25 reviewed sources. The deeper issue is that reported scores are *not directly comparable* across studies.

Scaling conventions alone create immediate confusion: CPG-LS reports on a 0.0-1.0 scale (achieving 0.9412) while other work uses 0-100 (with advanced white-box LLM samplers reaching 80-92). Hi-Stega's scores (0.1341-0.2051) look low by comparison, but actually represent nearly 10 $\times$ improvement over its own baseline (0.0135)-demonstrating that absolute values only matter within their own context.

Architectural differences further complicate matters: CPG-LS employs BERT-based lexical substitution whereas Hi-Stega uses generative GPT-2 models, making cross-study rankings invalid without careful normalization. Dataset choice compounds the problem-CPG-LS evaluated on CC-100 while Hi-Stega used Yahoo! News comments.

Like comparing temperatures without knowing Celsius from Fahrenheit, a "30" only makes sense in its original context. Consequently, MAUVE scores work best as *internal benchmarks* for comparing variants within a single study, not as universal performance indicators across different steganographic frameworks.

5.3.3 Statistical Metrics. Kullback-Leibler Divergence (KLD) [50] and Jensen-Shannon Divergence (JSD) are information-theoretic metrics used to evaluate steganographic security. KLD quantifies information loss by measuring the relative entropy between cover and stego distributions, serving as the theoretical standard for security modeling despite being asymmetric and failing as a strict distance measure. JSD improves upon this as a symmetric, bounded variant that measures how far each distribution lies from their average, providing a more stable basis for formulating statistical imperceptibility bounds-particularly when language models approximate human text distributions. Together, these two attempt to capture how closely steganographic outputs mimic legitimate communication channels.

However, real-world application reveals critical reliability failures, most notably the Perceptual-Statistical Imperceptibility Conflict (PSIC effect). KLD and JSD scores increasingly diverge from human judgment as statistical optimization progresses: methods achieving superior divergence metrics often produce chaotic, low-quality text easily detected by human observers. This discrepancy manifests acutely in dataset dependency-identical methods yield KLDs of 19.507 on IMDB versus 8.295 on Twitter at equivalent embedding rates, rendering cross-paper comparisons meaningless. Further compounding this, researchers employ incompatible formulas (some using latent BERT features versus direct word distributions), feature spaces, and measurement scales, evidenced by Meteor's KLD ranging from 0.045 in one study to 7.491-11.845 in others. Consequently, these metrics function like rulers measuring paintings: they confirm

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technical dimensional accuracy while completely missing perceptual naturalness, necessitating parallel evaluation with human-centric measures to achieve genuine security.

5.3.4 *Capacity Metrics.* Capacity is judged by four metrics:

- **Bits per Token (BPT):**

$$\text{BPT} = \frac{\text{Total Secret Bits}}{\text{Total Tokens}} \quad (1)$$

- **Bits per Word (BPW):**

$$\text{BPW} = \frac{\text{Total Secret Bits}}{\text{Total Words}} \quad (2)$$

- **Embedding Rate (ER) [51]:** Average density of hidden information per textual unit

$$\text{ER} = \frac{1}{N} \sum_{i=1}^N \text{bits}_i \quad (3)$$

where N is the number of textual units (words, tokens, or sentences) and bits_i is the number of bits embedded in the i -th unit.

- **Utilization Rate:**

$$H = - \sum_{x \in \mathcal{X}} P(x) \log_2 P(x) \quad (4)$$

$$\text{UR} = \left(\frac{\text{Actual Bits Embedded}}{H} \right) \times 100\% \quad (5)$$

These quantities quantify how densely a secret is packed, yet they are riddled with systematic biases that invalidate cross-system comparison.

Bias Category	Core Problem	Critical Implication
Tokenization Inconsistencies	Metrics depend entirely on specific tokenizers (e.g., GPT-2 BPE vs. word-level)	Direct comparisons across papers become meaningless when tokenization strategies differ
The "PSIC Effect"	Conflicts between imperceptibility and statistical security are ignored	High capacity may degrade human fluency while paradoxically improving detection resistance
Model Training Bias	Utilization Rate calculations assume uniform token availability	Actual hiding space is smaller than theoretical entropy due to model frequency preferences
Reporting Ambiguities	No standard definition of "capacity" across systems	Practice payload vs. effective payload distinctions create misleading efficiency claims
Context Blindness	Density metrics treat text as neutral bit containers	Semantic incoherence constitutes a security failure that BPW/BPT fails to penalize

Table 4. Five primary bias categories affecting capacity metrics in steganographic evaluation

Tokenization differences make "1 BPT" from one paper incomparable to "1 BPT" from another due to the use of different tokenizers. The PSIC effect shows that higher density can hurt human fluency yet help statistical evasion. Model frequency preferences shrink the real alphabet to high-probability tokens, so naive entropy limits overstate usable space. Ambiguous reporting-practice vs. effective payload, ER1 vs. ER2, Bit Length vs. Stego Length-lets authors cherry-pick flattering numbers. Finally, BPW/BPT ignore semantics, rewarding gibberish that is obviously steganographic.

Recent works reveal additional distortions: loop-count overhead, dictionary-size caps, baseline-dependent "improvements," watermarking goals that invert the desired signal, conversational filler that dilutes BPW, and nonlinear ER curves that make any single threshold misleading.

5.3.5 *Practical Capacity Analysis.* To contextualize theoretical capacity metrics, Table 6 summarizes the embedding capacity across the reviewed methods using specific generated samples. Capacity calculations vary significantly based

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on the encoding mechanism. **Fixed-length mapping** techniques, such as **LLM-Stega** and **GCStego**, derive capacity from rigid keyword or token mappings—64 bits per sentence and 16 bits per word, respectively. **ParaLS** operates similarly but on a smaller scale, encoding a simple 3-bit binary watermark (“110”).

In contrast, **payload-dependent** methods scale with the cover text or secret message length. **Meteor’s** capacity of 1280 bits corresponds to an embedded payload of 160 bytes (e.g., Lorem Ipsum text). **Zero-shot** capacity is directly tied to the Base64 representation of the secret bitstream (encoding 24 bits per 4 characters). Finally, **Discop** utilizes a statistical approach; with a truncation parameter of $p = 0.95$, it achieves an entropy rate of 4.84 bits per token, yielding 968 bits over a 200-token sequence.

Table 5. Examples of stego texts. Limited by the space, we omitted the padding part.

24-bit Secret Message : (1,0,0,0,1,0,0,1,0,1,1,0,1,0,1,0,1,1,1,1,1,0,1,1)		
Cover Text	Method	Stego Text
sometimes we bring the story to you, sometimes you have to go to the story.	Rewriting-Stego $BPT_S = 1$	sometimes we bring it to your mind but sometimes you have to go back to it to find the story you want to tell...
	Rewriting-Stego $BPT_S = 2$	but we have a different way of thinking about what the story...
	Rewriting-Stego $BPT_S = 4$	some of all the story lines...
	Bins $BPT_S = 4$	somewhere and we have lots different...
	Masked-Stega ($BPT_S = 0.16$) (Encrypted message :10001)	sometimes we read the story to you , sometimes you have to stick to the story...

[Placeholder footnote]

Table 6 provides a detailed catalog of stego text samples generated by the reviewed frameworks. It serves as a practical reference for the diverse capacities and calculation methods discussed previously, offering a direct comparison between the textual output and the numerical efficiency claimed by each study. This breakdown highlights the heterogeneity in how different researchers quantify the embedding density and its relation to the length of the generated cover.

Stego Text	Cap	Source	Calculation Basis
“The picture in The Pale IHCR scientists’ discussion now spans three dimensions. The first importance of the Yucatan Peninsula is demonstrated with the following conclusion: the Pliocene Earth has lost about seven times as much vegetation as the Jurassic in regular parts of the globe, from northern India to Siberia. Even since 1976, the continent has received or is already experiencing the worst mass extinction in recorded history since there were a million years last left from the extinction of Antarctic ice and the rapid spread of hydrocarbon-rich water as the Earth entered the Cretaceous Period. In turn, the world’s oceans have been breathtakingly transformed and as a result the surrounding environments are especially vulnerable. The reason is obvious: there was no geological record of the presence of the Yucatan Peninsula in the Late Miocene epoch, which is the line that runs from the southern edge of South America to Siberia. The commander-in-chief of the Yucatan Peninsula, Frederick Nystrom, has determined that the area is to stay as barren as possible. As a result, individuals living on lake beds and on coastal areas have witnessed the loss of about 90 percent of their habitat. The Yucatan Peninsula consists of four zones, with two different habitats separated, each of which has experienced in-seam damage. In one zone, along the northern shore of Lake Shemal, the retreating Tarahumara Ocean has been melted into a deep, seafloor called Nova Ravine, which south-east of the Yucatan Peninsula flows into Lake Isthmus, where there is an abundance of turtle life. A second beach, which lies at the far end of the peninsula, has been spewed down by a sea wall supporting Madagascar’s Great Ocean Earthquake, 9,000 feet in magnitude and caused large numbers of deaths. The third zone, along the coast of Cancun and in Asuncion, is less severely affected. An estimated 16 percent of the continent is protected from the destruction of oceanic winds and floods. All three zones are in a state of catastrophic destruction. According to the definition provided by the National Commission on the extinction of the dinosaurs (Infection and Immunization in the Ind”	1280 bits	Meteor	160-byte payload × 8 bits
“Washington was martyred in the battle of Cullman in 1788. Although down 19 men, yet his mission accomplished. He ended his life in 1896. When he died of disease, three of the men responsible said that Washington had been moved and was seen at the funeral placing him in his father’s arms. Another said that Washington treated his wounds with “a study of a personal sense of injustice.” (A 16th century German historian calls Washington a liar because of this website.) He was also a friend of Dr. Henry H. Jackson, who was notable as George Washington’s physician and a collaborator, and recognizes him by name in the any books about him. The last surviving manuscript is from 1888 and contains a frank and truthful account of the Quakers’ plight. One story states that while fighting in Whitesburg, Washington succumbed to pneumonia. He was 38 years old and according to a manuscript he got out the following year reports he grew old and fell in love. He also mentions a meeting with a woman who broke into his home and first went with him into a bath and gave him food and sleep. Three days later the woman left the room expecting him to eat her lunch and on that day he left home at 9:30 am in despair. He had not been to his bedside. On seeing this, he said a voice in him called out, “Your name is Jack. What is the girl?” Hamilton said the superior told him, “She was a layover in a bed and seven[Pg 209] feet below the bed where the general slept in very feminine attire. Nobody had time to look into her face. What was she to tell you about the general?” A Washington’s Official Address to Congress with Americans May 17th, 1781 “I am the one to announce completely that I am a true Christian and an eloquent philosopher. I am not constrained”	1280 bits	Meteor	160-byte payload × 8 bits
“Acclaimed dancer, Luna Moves, unassumingly announced her new song, which intriguingly entwines rhythmic beats with her signature performative flair.”	64 bits	LLM-Stega	64 bits/sentence (Subject, Predicate, Object, Emotion)
“The band’s praise of their latest song was received with neutrality, a calm before the potential storm of fan reactions.”	64 bits	LLM-Stega	64 bits/sentence
“The beloved band joyfully released an immersive exhibition, transporting fans through their rich musical journey, sparking waves of nostalgic happiness across the community.”	64 bits	LLM-Stega	64 bits/sentence

[Placeholder footnote]

Stego Text	Cap	Source	Calculation Basis
"The innovative game developers ecstatically released a gripping new episode, thrilling gamers and inviting them into captivating new worlds."	64 bits	LLM-Stega	64 bits/sentence
"With a voice that touched souls, the singer performed a memorable episode on the live show, casting a warm, positive light across the enchanted audience."	64 bits	LLM-Stega	64 bits/sentence
"I was genuinely underwhelmed by the film, and I'm afraid the only thing that stood out to me was the final scene. I had been eagerly anticipating it, but unfortunately the rest of the movie fell flat for me."	72 bits	Zero-shot	Base64 secret length (12 chars)
"All countries have an ugly side, just as ours does. It's unfair to say that only the Philippines has this issue, though. We are all capable of such shortfalls."	96 bits	Zero-shot	Base64 secret length (16 chars)
"Cheaper doesn't always have to equal inferior, particularly in the case of indie films, where creators are often pushed to maximize the impact of limited resources."	96 bits	Zero-shot	Base64 secret length (16 chars)
"Oh my gawd, stop dting married men! Like seriously, why do we keep falling into this trap over and over? Anglea basset's chracter should totall be more concerned with the fact that her husband abandoned hre for another woma than some mediocore romance flic. Overall, I would give this movie a big fat C- for wasted poasibility and a little more thought."	168 bits	Zero-shot	Base64 secret length (28 chars)
"Yes, I miss this cute kitty very much."	64 bits	GCStego	4 secret words $\times$ 16 bits/word
"A good subject to start."	3 bits	ParaLS	Explicit binary watermark "110"
"I was very nervous, at the time. I was 22, and the director was actually at a birthday party in Paris and just saying the doors were closed. But I put on a large black motorcycle, got my album saved and drove on, doing videos about history and the films. So I was really excited. And I found that in this movie, we were at a concert in London. And that concert was where every song and every act was at the movie playing the music that could be heard and the message was, "By God, there's people. But time has passed, and we're here, maybe from another world. Let's see how things can go." And at the end of the movie I said, "I hope the thing happened a bit, because there's no way you can change what you know. You know? Something has happened. Just let it happen. Let it happen." And at the end, I say to myself, "Hey, maybe we can change"	968 bits	Discop	200 tokens $\times$ 4.84 bits/token ($p = 0.95$)

Table 6. Practical embedding capacity and calculation basis for selected stego text samples.

5.3.6 *Security Metrics.* The sources evaluate security and reliability through the following integrated metrics:

- **Detection Accuracy (Acc) / Error Rate (PE):** Measures a classifier's ability to distinguish between cover and stego text. An accuracy of **50%** (or PE of 50%) is the gold standard, indicating the stego text is statistically identical to normal text.
- **F1 Score:** The harmonic mean of precision and recall, used to verify detection reliability, especially when datasets are balanced.
- **Robustness (Attack Resistance):** Evaluated via the **Mean Impact of Attack (mIOA)** or **Bit Accuracy** after removal attempts such as DAE (Denoising Auto-Encoder), word substitution, or sentence deletion.
- **False Positive Rate (FPR):** Measures how often human or normal model-generated texts are incorrectly flagged as containing secret messages.

5.3.7 *Evaluation Challenges and Gaps.* Several significant challenges exist in current evaluation practices:

- **Lack of Standardized Benchmarks:** The absence of shared benchmarks means only 20% of studies use common datasets, making comparison difficult
- **Inconsistent Reporting:** Different units, scales, and methodologies across studies
- **Limited Human Evaluation:** Only 25% of studies include human assessment

[Placeholder footnote]

- **Missing Robustness Testing:** 60% of studies don’t test against various attacks
- **Incomplete Evaluation:** Many studies focus on only one or two metric categories

5.4 Integration of External Knowledge Sources (RQ4)

The integration of external knowledge sources has emerged as a recurring design pattern in LLM-based steganography. Across the reviewed studies, external information is used primarily to improve contextual plausibility, maintain topic consistency, and in some cases expand the effective space available for embedding.

Knowledge Type	Typical Role	Primary Benefit	Examples
Semantic Resources	Topic and entity guidance	Better semantic control	Co-Stega, knowledge-graph guided methods
Domain Corpora	Domain adaptation	Better stylistic fit	FREmax, specialized datasets
Prompt Engineering	In-context control	Better task alignment	Zero-shot methods
Context Retrieval	Evidence or example retrieval	Higher contextual plausibility	Co-Stega, retrieval-augmented pipelines

Table 7. Representative patterns of external knowledge integration

5.4.1 *Semantic Resources Integration.* Rather than relying only on the internal prior of the language model, several recent systems inject external guidance during generation. Retrieved posts, headlines, or comments can provide a realistic discourse context for the cover text [33]. Knowledge-graph triples can constrain the content that should appear in the output, improving semantic consistency over longer spans [35]. Frequency-based external statistics can also be used to steer generation toward distributions that better match the target channel [34]. Taken together, these strategies show that contextual conditioning has become an important mechanism for balancing plausibility, control, and payload.

5.4.2 *Domain Corpora Integration.* Another recurring pattern is the use of domain-specific corpora. Fine-tuning or conditioning on domain data helps align the generated text with the vocabulary, stylistic conventions, and topical expectations of the intended communication channel. This pattern appears in systems such as VAE-Stega [1], Meteor [2], Hi-Stega [33], FREmax [34], Rewriting-Stego [31], and Summarization-Stego [37]. Although the underlying architectures differ, these methods share the assumption that better alignment with a target domain improves both naturalness and practical usability.

When additional training is not possible, black-box methods move this domain adaptation into the prompt or retrieval stage. Zero-shot approaches provide examples directly in the context window, LLM-Stega frames the generation request with topic-constrained prompts, and Co-Stega retrieves recent posts to anchor the generated reply in a plausible local context [15–17]. In these settings, external context functions as a lightweight substitute for model adaptation.

5.5 Limitations and Trade-offs in Current Techniques (RQ5)

This subsection consolidates the practical limitations previously introduced in the background discussion and reframes them as findings from the reviewed literature. Across the surveyed studies, the main constraints recur around imperceptibility, effective payload, semantic coherence, tokenization reliability, deployment assumptions, and attack robustness.

5.5.1 *The Perceptual-Statistical Imperceptibility Conflict (PSIC Effect).* Generative steganography faces an intrinsic tension: optimizing for perceptual fidelity degrades statistical security. Aggressive perplexity (PPL) minimization

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produces unnaturally “sharp” distributions that deviate from high-variance human writing, increasing detection vulnerability. Higher embedding rates force selection of lower-probability tokens, degrading text quality but expanding the candidate pool to enhance anti-steganalysis resistance. At elevated capacities, output distributions asymptotically approach natural language’s high-entropy chaos, camouflaging signals within legitimate statistical noise [1].

5.5.2 Attack Vulnerability and Security Concerns. Current techniques exhibit a substantial attack surface across multiple vectors, including word-level deletions and lexical substitutions that disrupt dependency parsing. Polishing and re-translation attacks progressively erase watermarks; at high intensities, these structural changes can reduce detection performance to an F1 score of 0.5, rendering the watermark undetectable. Furthermore, adversarial machine learning enables the extraction of secret model parameters from black-box systems, potentially compromising integrity entirely. To mitigate these risks, Variational Auto-Encoders (VAE) are employed to learn the statistical distribution of normal text, aiming to reduce the KL divergence between cover and stego distributions and resolve the conflict between perceptual and statistical imperceptibility [45].

5.5.3 Capacity and Contextual Constraints. Short, low-entropy contexts severely constrain embedding capacity. Social media posts and technical documents offer minimal redundancy for hiding [16], while semantic preservation requirements compete directly with information embedding objectives. Brief texts further lack sufficient contextual support for effective steganographic operations [6].

Early generative methods also produced fluent but semantically arbitrary text, violating cognitive imperceptibility [6]. When generation is conditioned only on local token history, the output may drift away from the source topic or conversational logic, which is especially suspicious in low-entropy channels such as social media replies. This contextual drift worsens when methods must keep generating until an arbitrarily large payload is fully embedded: texts become unnaturally long, coherence decays over time, and practical payload efficiency drops even when reported bit capacity appears high [32, 35, 40, 44].

5.5.4 Segmentation and Tokenization Barriers. Subword tokenization introduces critical extraction ambiguities that are **fundamental to white-box steganographic methods**. Byte-pair encoding splits words unpredictably, creating multiple valid segmentations that corrupt message boundaries and cause decoding failures. These ambiguities impose hard capacity caps regardless of algorithmic sophistication—**SparSamp** suffers accuracy degradation from token ambiguity (TA), while **ShiMer** cannot effectively boost entropy due to tokenization constraints [21].

The core issue lies in **segmentation and tokenization barriers**—also termed **segmentation ambiguity**—which persist **even when sender and receiver employ identical Large Language Models and tokenizers**. Because modern LLMs decompose words into subword units, and vocabularies frequently contain both complete words and their constituent components, any given text string admits multiple valid token representations. This multiplicity is compounded by the **detokenization-retokenization gap**: senders must convert tokens to plain text to conceal their transmission, while receivers must subsequently retokenize this text to recover hidden messages. Despite shared tokenization rules, receivers may select different token sequences than senders intended—senders might encode messages using separate tokens (“any” and “thing”), only for receivers to retokenize “anything” as a **single token** (“anything”). Because LLMs operate **autoregressively**, a single divergent token choice alters the probability distribution for all subsequent predictions, causing **total decoding failure** for the remainder of the message. To address this **white-box-specific challenge**, researchers propose **SyncPool**, a mechanism that aggregates ambiguous tokens into

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pools and employs synchronized random number generators to ensure both parties select **identical tokens**, thereby eliminating ambiguity and guaranteeing reliable message extraction. [21]

5.5.5 White-box versus Black-box Trade-offs. The architectural choice between white-box and black-box access paradigms involves fundamental trade-offs. White-box methods require shared language models and training vocabularies, achieving statistical imperceptibility through distribution learning (e.g., VAE-Stega), but suffer from low accessibility and deployment complexity. Black-box alternatives (e.g., LLM-Stega) offer high accessibility via standard user interfaces with capacity of 5.93 bpw and PPL of 165.76, yet still exhibit the PSIC effect, where quality and security remain in tension. These dimensions are orthogonal to methodological hybrids: **Hi-Stega**[33] combines retrieval-based and generation-based paradigms; **Rewriting-Stego**[31] merges edit-based and generation-based approaches using a denoising Seq2Seq model (BART); **Joint Linguistic Steganography**[38] integrates conditional generation with BERT-based substitution and hybrid temporal-spatial feature extraction via Graph Attention Networks; **Co-Stega**[16] fuses efficient retrieval with generative replies; and **Patient-Arithmetic**[6] in NMT-Stega hybridizes arithmetic coding with waiting mechanisms to avoid probabilistic cliffs during translation.

5.5.6 Computational and Resource Constraints. Performance optimization conflicts directly with computational efficiency [2]. Superior results demand increased resources and memory, particularly for large models with external knowledge bases [20]. Real-time latency constraints limit optimization depth, while performance degradation emerges at operational scale. Variable length sampling and rejection sampling significantly increase encoding times, and LLM-based steganography requires 3x to 5x more time than prior methods [21]. The **Meteor** system exemplifies hardware constraints, running nearly 50× slower on mobile devices compared to GPU environments.

LLM-specific deployment obstacles compound these costs. Hallucinations and nonsensical continuations can corrupt the covert bitstream or create detectable artifacts, while lossy transmission, editing, or platform-side normalization can irreversibly damage extraction. As model quality improves, steganalysis systems also improve, producing an arms race in which better generation does not automatically translate into better security.

Black-Box State Considerations. Most users operate in a **black-box state**, accessing LLMs solely through APIs or UIs without internal visibility. This restriction denies access to training vocabulary and sampling probabilities essential for traditional white-box steganographic mapping. Consequently, researchers have developed **black-box generative steganography** (e.g., LLM-Stega) utilizing prompts and keyword sets rather than direct logit manipulation. The black-box nature has also shifted watermarking toward **post-hoc injection**, embedding marks into pre-generated text since the generation process itself remains unmodifiable by end users.

5.5.7 Critical Unresolved Challenges. Foundational gaps persist across the research landscape: theoretical security guarantees remain unestablished, resilience to advanced attacks is limited, evaluation frameworks lack standardization, responsible development guidelines are absent, non-English performance lags substantially, and real-world deployment testing remains insufficient. Several studies also surface unresolved concerns around reversibility, bias, and misuse potential, indicating that robust engineering alone will not be enough without clearer security and governance models [17, 36, 41, 52].

5.5.8 Quantified Impact Summary. Quantitative evidence highlights several constraints on current steganographic systems. The Perceptual-Statistical Imperceptibility Conflict (PSIC) effect creates a trade-off where increasing the embedding rate (e.g., to 5 bpw) strengthens anti-steganalysis but significantly degrades perceptual text quality [1].

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Attack vulnerability is notable; for example, adaptive attacks can reduce bit accuracy to near-random levels (0.6) [39]. Segmentation ambiguity and entropy variability in communication channels are major impediments that lead to decoding failures and restricted practical capacity [2]. Finally, computational optimizations induce significant overhead, with some reordering algorithms causing up to 69% performance degradation in processing speed [20].

6 DISCUSSION

This section synthesizes the findings across the five research questions and discusses how recent LLM-based approaches are changing the design priorities of linguistic steganography.

6.1 The Emergence of Meaning-Based Steganography

One of the clearest patterns in the reviewed studies is the shift from strictly white-box, probability-driven methods toward broader black-box and context-aware approaches. Earlier linguistic steganography often depended on local lexical or sampling manipulations that were difficult to scale and easy to distort semantically. More recent work uses LLMs to support full-text generation, rewriting, prompting, and retrieval, thereby moving the problem from isolated token selection toward broader control of discourse and context.

6.2 Resolving the PSIC Effect through Contextual Compatibility

The Perceptual-Statistical Imperceptibility Conflict (PSIC) remains one of the main technical hurdles identified across the reviewed studies. The synthesis suggests that **contextual compatibility** is not simply an optional enhancement, but an increasingly important strategy for mitigating this conflict.

By integrating external knowledge sources (RQ4), recent systems can better align generated text with the topic, style, and discourse expectations of a target channel. In practical terms, this means that security is increasingly tied not only to distributional similarity, but also to whether the text remains semantically plausible within a recognizable communicative setting.

6.3 Implications for Research and Practice

6.3.1 The Democratization of Covertress. The shift toward black-box approaches (RQ1, RQ2) lowers the practical barrier to deployment. By treating models as opaque APIs or hosted services, covert communication is no longer limited to settings where both parties can share and run the same model locally. This may improve accessibility in constrained environments, but as noted in RQ5 it also lowers the barrier for misuse.

6.3.2 The Need for Semantic-First Evaluation. The evaluation metrics reviewed in RQ3 show that the field still relies heavily on surface-level or model-dependent metrics such as perplexity and bit-density measures, while human and semantic evaluation remain less common. This imbalance makes it difficult to assess whether apparently strong results also preserve coherence, relevance, and contextual plausibility. Future benchmarking should therefore place more emphasis on semantic and human-centered evaluation alongside statistical measures.

6.4 Ethical Responsibility and the Forensics Arms Race

The reviewed literature also highlights a gap between technical innovation and explicit discussion of misuse, governance, and defensive implications. As LLM-based steganography becomes more fluent and easier to deploy, steganalysis and provenance research will likely need to adapt beyond traditional lexical or shallow statistical features.

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6.5 Limitations of the Review

This review should be interpreted as a structured snapshot of a rapidly changing field. The pace of LLM development, the heterogeneity of evaluation settings, and the exclusion of six records pending PDF acquisition all limit the stability of any quantitative summary.

6.6 Future Research Directions

Based on the identified gaps, three research directions appear especially important:

- **Provable Semantic Security:** Extending security analysis beyond token distributions toward stronger guarantees about contextual plausibility and recoverability.
- **Multimodal Grounding:** Studying how text-based hiding strategies interact with multimodal systems without collapsing the review boundary between steganography and adjacent security problems.
- **Adaptive Context Synthesis:** Designing retrieval and prompting pipelines that improve realism without making systems excessively fragile, slow, or difficult to evaluate.

7 CONCLUSION

This systematic literature review shows that Large Language Models (LLMs) have substantially expanded the design space of linguistic steganography. Across 25 primary studies, the reviewed literature points to a shift toward generative, rewriting-based, black-box, and context-aware techniques that aim to improve fluency and contextual plausibility while maintaining usable payload.

At the same time, the review shows that these advances do not remove the core constraints of the field. Evaluation remains inconsistent, comparisons across studies are often difficult, and persistent trade-offs remain among imperceptibility, capacity, robustness, and deployability. In particular, the Perceptual-Statistical Imperceptibility Conflict (PSIC), low-entropy communication settings, tokenization ambiguity, and black-box limitations remain central obstacles.

One of the clearest themes across the reviewed studies is the growing importance of contextual compatibility. Domain adaptation, retrieval, and semantic guidance are increasingly used to make stego text more plausible within a target communicative setting. This suggests that future progress will depend not only on better language models, but also on stronger evaluation protocols and more disciplined integration of context.

Future work should therefore focus on standardized evaluation, more reliable extraction under realistic tokenization and editing conditions, stronger security guarantees for practical systems, and clearer boundaries between linguistic steganography and adjacent areas such as watermarking or broader AI security. Overall, the literature suggests that LLMs have made linguistic steganography more practical and expressive, but not yet methodologically mature.

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Table 8. Summary of Results from Reviewed Papers

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
VAE-Stega: linguistic steganography based on va... [1]	BERTBASE (BERT-LSTM) (LSTM-LSTM) model was trained from scratch	2020.0	Twitter (2.6M sentences) IMDB (1.2M sentences) preprocessed	PPL: 28.879, Δ MP: 0.242, KLD: 3.302, JSD: 10.411, Acc: 0.600, R: 0.616	non-explicit	pre-text	text
General frame-work for re-versible data hiding in... [36]	BERTBase	2022.0	BookCorpus	BPW=0.5335 F1=0.9402 PPL=134.2199	non-explicit	pre-text	text
Co-stega: Collaborative linguistic steganograph... [16]	Llama-2-7B-chat, GPT-2 (fine-tuned), Llama-2-13B	2024.0	Tweet dataset (for GPT-2 fine-tuning), Twitter (real-time testing)	SR1: 60.87%, SR2: 98.55%, Gen. Capacity: 44.91 bits, Entropy: 49.21 bits, BPW: 2.31, PPL: 16.75, SimCSE: 0.69	explicit	Social Media	text

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Joint linguistic steganography with BERT masked... [38]	LSTM + attention for temporal context. GAT for spatial token relationships. BERT MLM for deep semantic context in substitution.	2023.0	OPUS	PPL=13.917 KLD=2.904 SIM=0.812 ER=0.365 (BN=2) Best Acc=0.575 (BERT classifier) FLOPs=1.834G	explicit	pre-text	text
Discop: Provably secure steganography in practi...	GPT-2	2023.0	IMDB	p=1.00 Total Time (seconds)=362.63 Ave Time ↓ (seconds/bit)=6.29E-03 Ave KLD ↓ (bits/token)=0 Max KLD ↓ (bits/token)=0 Capacity (bits/token)=5.76 E...	non-explicit	tuning + pretext	text

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Generative text steganography with large language... [15]	Any	2024.0	[Not specified]	Length: 13.333 (words). BPW: 5.93 bpw PPL: 165.76. Semantic Similarity (SS): 0.5881 LS-CNN Acc: 51.55%. BiLSTM-Dense Acc: 49.20%. Bert-FT Acc: 50...	explicit	[Not specified]	[Not specified]
Meteor: Cryptographically secure steganography ... [2]	GPT-2	2021.0	Hutter Prize, HTTP GET requests	GPT-2: 3.09 bits/token	non-explicit	tuning + pretext	text
Zero-shot generative linguistic steganography [17]	LLaMA2-Chat-7B (as the stegotext generator / QA model). GPT-2 (for NLS baseline and JSD evaluation)	2024.0	IMDB, Twitter	PPL: 8.81. JSDfull: 17.90 (x10[truncated]iicircum-2). JSDhalf: 16.86 (x10[truncated]iicircum-2). JSDzero: 13.40 (x10[truncated]iicircum-2) TS...	explicit	zero-shot + prompt	text

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Provably secure disambiguating neural linguisti... [21]	LLaMA2-7b (English), Baichuan2-7b (Chinese)	2024.0	IMDb dataset (100 texts/sample, 3 English sentences + Chinese translations)	Total Error: 0%, Ave KLD: 0, Max KLD: 0, Ave PPL: 3.19 (EN), 7.49 (ZH), Capacity: 1.03–3.05 bits/token, Utilization: 0.66–0.74, Ave Time: [truncat...	non-explicit	pretext	text
A principled approach to natural language water... [39]	Transformer-based encoder/decoder; BERT for distillation	2024.0	Web Transformer 2	Bit acc: 0.994 (K=None), 1.000 (DAE), 0.978 (Adaptive+K=S); Meteor Drop: [truncated]iitilde0.057; SBERT ↑: [truncated]iitilde1.227; Ownership R...	Yes; semantic-level embedding; synonym substitution using BERT	Yes; watermark message assigned categorical label (e.g., 4-bit → 1-of-16)	Yes; semantic embeddings via transformer encoder and BERT; SBERT distance as metric

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Context-aware linguistic steganography model ba... [6]	BERT (encoder), LSTM (decoder)	2024.0	WMT18 News Commentary (train/test), Yang et al. bits, Doc2Vec, 5,000 stego pairs (8:1:1 split)	BLEU: 30.5, PPL: 22.5, ER: 0.29, KL: 0.02, SIM: 0.86, Stego detection [truncated]iitilde16%	Yes	[Not specified]	GCF (global context), LMR (language model reference), Multi-head attention
DeepTextMark: a deep learning-driven text water... [42]	Model-independent; tested with OPT-2.7B	2024.0	Dolly ChatGPT (train/validate), C4 (test), robustness & sentence-level test sets	100% accuracy (multi-synonym, 10-sentence), mSMS: 0.9892, TPR: 0.83, FNR: 0.17, Detection: 0.00188s, Insertion: 0.27931s	NO	[Not specified]	[Not specified]
Hi-stega: A hierarchical linguistic steganograp... [33]	GPT-2	2024.0	Yahoo! News (titles, bodies, comments); 2,400 titles used	ppl: 109.60, MAUVE: 0.2051, ER2: 10.42, $\Delta(\text{cosine})$: 0.0088, $\Delta(\text{simcse})$: 0.0191	explicit	Social Media	Text

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Linguistic steganography: From symbolic space t... [28]	CTRL (generation), BERT (semantic classifier)	2020.0	5,000 CTRL-generated texts per semanteme (n = 2–16); 1,000 user-generated texts for anti-steganalysis	Classifier Accuracy: 0.9880; Loop Count: 1.0160; PPL: 13.9565; Anti-Steganalysis Accuracy: [truncated]	implicit	Text	Semanteme (α) as a vector in semantic spac
Natural language steganography by chatgpt [32]	[Not specified]	2024.0	Custom word sets for specific topics (e.g., 16×10-word sets for music reviews)	[Not specified]	Explicit	Specific Genre/Topic Text	Text
Natural language watermarking via paraphraser-b... [41]	Transformer (Paraphraser), BART (BARTScore), BERT (BLEURT, comparisons)	2023.0	ParaBank2, LS07, CoInCo, Novels, WikiText-2, IMDB, NgNews	LS07 P@1: 58.3, GAP: 65.1; CoInCo P@1: 62.6, GAP: 60.7; Text Recoverability: [truncated]	Explicit	[Not specified]	text

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Rewriting-Stego: generating natural and control... [31]	BART (bart-base2)	2023.0	Movie, News, Tweet	BPTS: 4.0, BPTC+S: 4.0, PPL: 62.1, Mean: 44.4, Variance: 2.1e04, Acc: 8.9%	not Explicit	[Not specified]	[Not specified]
ALiSa: Acrostic linguistic steganography based ... [19]	BERT (Google’s BERTBase, Uncased)	2022.0	BookCorpus (10,000 natural texts for evaluation)	PPL: Natural = 13.91, ALiSa = 14.85; LS-RNN/LS-BERT Acc & F1 = [truncated]iitilde0.50; Outperforms GPT-AC/ADG in all cases	No	[Not specified]	[Not specified]
Imperceptible Text Steganography based on Group...	Qwen-7B-Chat	2024.0	HC3, DailyDialogue, COCO Descriptions	HC3: Bit 188.94, Stego 131.99, PPL 34.07, Mean 20.19, Var 0.1e04, F1 90.01%; DailyDialogue: Bit 188.94, Stego 89.37, PPL 53.88, Mean 20.13, Var 0....	Explicit	Social Media / Group Chat	Text (chat history and current input)
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Table 8 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
A Semantic Controllable Long Text Steganography...	Llama 7B Chat, Meta LLaMA2 7B Chat	2024.0	Story (ChatGPT), Post (Recipe Kaggle + ChatGPT), Ad (Mobile Kaggle + ChatGPT)	ppl ↓ >23%, Appl ↓ >72% vs ADG/HC/Bin; detection accuracy ↓ >10% vs baselines	Explicit	Topical Content	KG triplets (e1, r, e2), task descriptions (D)
Beyond Binary Classification: Customizable Text...	gpt-3.5-turbo-instruct, OPT-6.7b, babbage-002, davinci-002 (others: ChatGPT, GPT-2-4, LLaMA)	2024.0	Realnewslike (C4, 500 samples, 100-token prompts + completions); Custom watermark dataset (short info <10 tokens)	AUC 0.98, FPR 0.00, FNR 0.00, [truncated]iitilde100% single-letter decoding, PPL close to human text	Implicit	General Text Generation	Text (evolving prompt + generated output)
CPG-LS: Causal Perception Guided Linguistic Ste...	BERTBase, Cased	2023.0	CC-100 corpus; 10k cover texts; 7:3 train-test split	PPL 36.5; Mauve 0.871; Payload 0.150 bits/word; BiLSTM-D Acc 0.387 F1 0.375; R-BI-C Acc 0.378 F1 0.366; TS-RNN Acc 0.380 F1 0.368	Implicit	Natural Language Text	Text, embeddings, vector matrix

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Table 8 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Controllable Semantic Linguistic Steganography ...	BERT + CRF	2024.0	Gigaword; CNN/Daily Mail	Rouge-1: 0.2212; Rouge-2: 0.0268; Rouge-L: 0.1609; Meteor: 0.1384; Cosine: 0.5911; Euclidean: 5.6386; Manhattan: 87.9534; Jaccard: 0.2022; Anti-ste...	Explicit	Social Media	Semantic features of input text; 384-dim dense vectors for evaluation
FREmax: A Simple Method Towards Truly Secure Ge...	GPT-2	2024.0	Tweet corpus (2.6M sents, 26.8M tokens), IMDB corpus (1.05M sents, 25.3M tokens)	Tweet: PPL 361.83, Entropy 48.21, Tokens 10.83, Distinct3 0.98, BPS 62.79, SI% 73.03. IMDB: PPL 169.66, Entropy 103.39, Tokens 23.80, Distinct3 0....	Implicit	General Text	N-gram frequency distribution stored in a look-up table

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CONTENTS

Abstract	1
1 Introduction	1
2 Background	2
2.1 Capabilities and Approximating Natural Communication	3
2.2 Role in Generative Linguistic Steganography	3
3 Related Reviews	4
4 Research Method	4
4.1 Planning	4
4.1.1 Research Questions	4
4.1.2 Search Strategies	5
4.1.3 Inclusion and Exclusion Criteria	5
4.2 Conducting the Search	5
4.3 Data Extraction and Classification	5
5 Results	6
5.1 State of Published Literature on LLM-based Steganography (RQ1)	6
5.1.1 Publication Trends and Distribution	6
5.2 Applications of LLM-based Steganographic Techniques (RQ2)	7
5.3 Evaluation Metrics and Methods (RQ3)	7
5.3.1 Perplexity (PPL)	7
5.3.2 MAUVE	8
5.3.3 Statistical Metrics	8
5.3.4 Capacity Metrics	9
5.3.5 Practical Capacity Analysis	9
5.3.6 Security Metrics	12
5.3.7 Evaluation Challenges and Gaps	12
5.4 Integration of External Knowledge Sources (RQ4)	13
5.4.1 Semantic Resources Integration	13
5.4.2 Domain Corpora Integration	13
5.5 Limitations and Trade-offs in Current Techniques (RQ5)	13
5.5.1 The Perceptual-Statistical Imperceptibility Conflict (PSIC Effect)	13
5.5.2 Attack Vulnerability and Security Concerns	14
5.5.3 Capacity and Contextual Constraints	14
5.5.4 Segmentation and Tokenization Barriers	14
5.5.5 White-box versus Black-box Trade-offs	15
5.5.6 Computational and Resource Constraints	15
5.5.7 Critical Unresolved Challenges	15
5.5.8 Quantified Impact Summary	15
6 Discussion	16
6.1 The Emergence of Meaning-Based Steganography	16

1410	6.2	Resolving the PSIC Effect through Contextual Compatibility	16
1411	6.3	Implications for Research and Practice	16
1412	6.3.1	The Democratization of Covertness	16
1413	6.3.2	The Need for Semantic-First Evaluation	16
1414	6.4	Ethical Responsibility and the Forensics Arms Race	16
1415	6.5	Limitations of the Review	17
1416	6.6	Future Research Directions	17
1417	7	Conclusion	17
1418		References	17
1419		Contents	29
1420			
1421			
1422			
1423			
1424			
1425			
1426			
1427			
1428			
1429			
1430			
1431			
1432			
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